

Marine A. Denolle

Associate Professor, Earth and Space Sciences, University of Washington | College of Environment CV — reverse chronological order; active/completed 2023–present appear first

📍 Seattle, WA ✉ mdenolle@uw.edu 🌐 www.denolle.ai 🐙 mdenolle 🏠 Denolle-Lab 📞 0000-0002-1610-2250

👤 Google Scholar 📺 scoped6259 📺 uwgeophysics6888 🏠 denolle-lab.github.io

Education

- 2008 – 2014 **Stanford University** Geophysics
 - Supervisor: Dr. Gregory Beroza
 - Co-supervisors: Dr. Eric Dunham, Dr. Jesse Lawrence
- 2007 – 2008 **Ecole Normale Supérieure - IPGP** Geophysics
 - Supervisors: Dr. Satish Singh (IPGP), Dr. David Bercovici (Yale)
- 2006 **Ecole Normale Supérieure** Earth Sciences
- 2004 – 2005 **Lycée Chateaubriand, Rennes** Physics-Mathematics

Research Impact Summary

Research Citations: 3,513 total (h-index: 25, i10: 47) | 2,422 last-5y (h₅: 23, i10₅: 45) | Google Scholar

Publications: 68 peer-reviewed | 5 Nature/Science-family (T1) | 40 AGU-flagship (T2) | 19 domain journals (T3) | 3 reviews/perspectives | 1 preprints

Research Funding: Lead PI: \$3.1M (14 grants) | Co-PI/Co-I: \$4.9M (8 grants) | Fellowships: \$878K (2)

Open-Source Software: 372 GitHub stars | 241 forks | 18 active repos | NoisePy: 207 stars, 83 forks, 19 contributors | 1,692+ PyPI downloads/yr

Mentoring: 8 PhD (4 current, 4 graduated) | 11 postdocs | 15+ undergrads | 10 co-supervised grad students | 5 became faculty

Broader Impact: 6 workshops, 215+ seismologists trained | mlgeo-book: 11 GitHub stars

Employment

- 2025 – present **Co-Founder**, Applied Environmental Intelligence – Seattle
- 2024 – present **Associate Professor**, University of Washington – Seattle, WA
Department of Earth and Space Sciences
- 2021 – 2024 **Assistant Professor**, University of Washington – Seattle, WA
Department of Earth and Space Sciences
- 2016 – 2021 **Assistant Professor**, Harvard University – Cambridge, MA
Department of Earth and Planetary Sciences
- 2014 – 2016 **Green Postdoctoral Fellow**, UC San Diego, Scripps Institution of Oceanography – La Jolla, CA
Institute of Geophysics and Planetary Physics
 - Supervisor Dr. Peter Shearer

Awards

- 2023 **Invited Professorship** Ecole Normale Supérieure rue d'Ulm-Paris

- 2023 **Data Science Fellow** eScience Institute, University of Washington
- 2019 **Charles F. Richter Early Career Award** Seismological Society of America
- 2019 **Kavli Frontiers of Science Fellow** National Academy of Sciences
- 2019 **Radcliffe Assistant Professorship** Institute for Advanced Study
- 2018 **CAREER Award** NSF
- 2017 **David and Lucile Packard Foundation Fellowship** David and Lucile Packard Foundation
- 2016 **Outstanding Reviewer** Geophysical Research Letters
- 2015 **Outstanding Reviewer** Geophysical Journal International
- 2012 **AGU Outstanding Student Paper Award** American Geophysical Union
- 2012 **SSA Student Presentation Award** Seismological Society of America
- 2010 **AGU Outstanding Student Paper Award** American Geophysical Union

Teaching

- 2023 - 2025 **Computational/Advanced Seismology**, ESS 563 – U of Washington
- Taught Spring 2023, 2024, 2025
 - Covers ambient noise seismology, earthquake source characterization
 - Julia and Python programming for high-performance computing
- 2023 - 2026 **Introduction to Seismology**, ESS 412/512 – U of Washington
- Taught Winter 2023, 2025, 2026
 - Covers earthquake physics, seismic waves, and hazard assessment
- 2022 - 2026 **Geophysics**, ESS 314 – U of Washington
- Taught 2022, 2023, 2024, 2026
- 2021 - 2024 **Machine Learning in the Geosciences**, ESS 469/569 – U of Washington
- Taught Fall 2021, 2022, 2023, 2024
 - Open-access Jupyter Book with asynchronous teaching materials
 - Course adopted by University of Arizona and UC Berkeley
 - Curated geoscience-specific ML datasets for homework
- 2019 **Machine Learning in Earth and Planetary Sciences**, EPS 268 – Harvard
- Taught Fall 2019
- 2018 **Induced Seismicity**, EPS 268 – Harvard
- Taught Fall 2018
- 2018 - 2020 **Earthquakes and Faulting**, EPS 203 – Harvard
- Taught Fall 2018, 2020
- 2017 - 2020 **Earthquakes and Tectonics**, EPS 55 – Harvard
- Taught Fall 2017, 2020

2016 **Earthquake Sources**, EPS 204 – Harvard

- Taught Spring 2016

Phd Advisees

2025 - present **Michael Hemmett**, PhD Pre-Candidate – UW ESS

Offshore geophysics

2022 - present **Manuela Koepfli**, PhD Candidate – USS

Geohazard assessment and early warning systems

- 1 publication, 1 in review, 1 in prep

2022 - present **Akash Kharita**, PhD Pre-Candidate – UW ESS

Geohazard characterization using machine learning

- 2 publications

2021 - present **Yiyu Ni**, PhD Candidate – UW ESS

Machine learning and big data seismology

- 8 publications

2019 - 2024 **Congcong Yuan**, PhD – Harvard

Time-dependent seismology, solid-fluid interaction. Next position: Cornell postdoc → Faculty position at NTU Singapore

- 4 publications

2018 - 2023 **Stephanie Olinger**, PhD – Harvard

Cryo-seismology (50% co-advised with Brad Lipovsky). Next position: Stanford Thompson Postdoctoral Fellowship → CEO of AEI

- 4 publications

2016 - 2021 **Tim Clements**, PhD – Harvard

Hydro-seismology and big-data seismology. Next position: USGS Mendenhall Postdoc → USGS Research Geophysicist

- 4 publications

2016 - 2022 **Jiuxun Yin**, PhD – Harvard

Earthquake seismology and source characterization. Next position: Caltech SCSN Postdoc Fellowship → Schlumberger

- 6 publications

Postdocs

2024 - 2025 **Dr. Stephanie Olinger**, University of Washington

Cryoseismology, DAS, Ambient noise monitoring

- Next position: Climate Tech → CEO Applied Environmental Intelligence

2023 - 2024 **Dr. Kuan-Fu Feng**, University of Washington

Environmental Seismology, Cloud computing

- Next position: Postdoc at University of Utah

2023 - 2025 **Dr. Ethan Williams**, University of Washington

DAS

- Next position: Assistant Professor at UC Santa Cruz

- 2022 - 2024 **Dr. Qibin Shi**, University of Washington
DAS, Deep Learning, Earthquake Catalog building, AgroSeismology
• Now Postdoctoral Fellow at Rice University
- 2020 - 2022 **Dr. Laura Ermert**, Harvard University/University of Washington
Environmental seismology
• Next position: Assistant Professor at ISTerre
- 2019 - 2020 **Dr. Xiaotao Yang**, Harvard University
Ambient noise cross correlation
• Next position: Assistant Professor at Purdue
- 2019 - 2020 **Dr. Kurama Okubo**, Harvard University
Ambient noise monitoring
• Next position: Researcher at NIED, Japan
- 2019 - 2020 **Dr. Zhitu Ma**, Harvard University
Ambient noise cross correlation
• Next position: Assistant Professor at Tongji University, China
- 2018 - 2019 **Dr. Chengxin Jiang**, Harvard University
Ambient noise seismology, computational seismology
• Next position: Research Associate at Australian National University
- 2016 - 2017 **Dr. Chris Van Houtte**, Harvard University
Earthquake Source seismology
- 2016 - 2018 **Dr. Loïc Viens**, Harvard University
ambient noise seismology, ground motion prediction
• Next position Researcher at Los Alamos

Other Graduate Student Supervision

Note: (*) At UW, roles are secondary advisor or primary advisor on one project/manuscript. (**) Advising resulted in a peer-reviewed publication.

Maleen Kidiwela (*, **): 2021-present, UW Oceanography (co-advised with William Wilcock) - DAS and offshore seismology, resulted in publication

Zoe Krauss (*, **): 2021-2024, UW Oceanography (co-advised with William Wilcock) - Axial Seamount hydrothermal monitoring, resulted in publication

Parker Sprinkle (*, **): 2021-present, UW ESS (co-advised) - Earthquake early warning and machine learning

William Flanagan (*, **): 2019, Harvard EPS - Volcano seismology project

Congcong Yuan (*, **): 2019, USTC China - Master student visiting researcher on time-dependent seismology, resulted in publication

Natasha Toghramadjian (*, **): 2018-2025, Harvard EPS - Seismic attenuation and site effects, resulted in publications

Zhuo Yang (*, **): 2018-2021, Harvard EPS - Machine learning for seismic phase picking, resulted in publication

Philippe Danré (*, **): 2018, Ecole Normale Supérieure Paris - Master thesis on seismic monitoring, resulted in publication

Thibault Pérol (*, **): 2017, Harvard EPS - Deep learning for earthquake detection, resulted in multiple publications

Manuel Florez (*, **): 2017-2019, MIT - Dissertation committee member for geophysics PhD

Graduate Student Committees

Jones, Randall: Atmospheric Sciences, GSR, 2025-present

Koepfli, Manuela: Earth and Space Sciences, Chair, 2024-present

Ni, Yiyu: Earth and Space Sciences, Chair, 2024-present

Pearson, Anna Elaine Rogers: Earth and Space Sciences, Member, 2024-present

Kidiwela, Maleen: School of Oceanography, Member, 2024-present

Sangmin, Song: School of Oceanography, GSR, 2024-present

Sweeney, Aodhan: Atmospheric Sciences, GSR, 2024-present

Kharita, Akash: Earth and Space Sciences, Advisor, 2023-present

DeGrande, Jensen: Earth and Space Sciences, Member, 2023-present

Zhang, Maochuan: School of Oceanography, GSR, 2023-present

Chien, Mu-Ting: Atmospheric Sciences, GSR, 2023-2024

Ragland, John: Electrical and Computer Engineering, GSR, 2023-2024

Velappan, Hemalatha: School of Environmental and Forest Sciences, GSR, 2023-present

Velegar, Meghana S: Applied Mathematics, GSR, 2023-2023

Sprinkle, Parker: Earth and Space Sciences, Member, 2022-present

Krauss, Zoe: School of Oceanography, GSR, 2022-2024

Rasanen, Ryan: Civil and Environmental Engineering, GSR, 2022-2023

Zahn, Olivia: Physics, GSR, 2022-2024

International Phd Advising

Note: Dissertation and defense evaluating committee members

Marius Paul Isken : 2024 (GFZ, Germany)

Luc Illien : 2023 (GFZ, Germany)

Zoe Renate : 2023 (Universite de Lorraine, France)

Reza Esfahani : 2022 (GFZ, Germany)

Kurama Okubo : 2019 (ENS, Paris, France)

Media Coverage

- [2025 Global-scale Database of Seismic Phases](#)
- [2024 Ambient Field Monitoring in the Shallow Crust](#)
- [2024 Making Big Data Look Small](#)
- [2023 UW eScience Seminar](#)
- [2023 Open Science in Seismology](#)
- [2023 UW News Fiber Sensing](#)
- [2018 Radcliffe Seminar: Where we stand in Earthquake Prediction](#)

Undergraduate Students

Note: My undergraduate advising includes research opportunities through summer programs and independent studies (ESS 499 for credit during academic year, hourly pay during summer/breaks). I mentor students toward conference presentations and publications, and support competitive fellowship applications. Legend: (*) Conference presentation | (**) Peer-reviewed publication | (***) Publication in prep | (+) NSF GRFP recipient

Alex Rose : 2025-present, UW Oceanography - DAS data processing research

Anjani Mirchandi: 2024-present, UW Applied Math - DAS data processing research

Hiroto Bito: 2023-2026, UW ESS - ML for earthquake catalog building and spatio-temporal forecasting

Nicholas Wolfe: 2023, UW ESS - Earthquake magnitude estimation project

Informatics Capstone Team (lead Rona Guo*): 2023, Nathan Limono, William Phan, Michael Yung, Matthew Herradura, UW Informatics - DAS Platform development, resulted in conference presentation

Lucas Swanson : 2023, UW Informatics - DAS software development

Francesca Skene (*) : 2022-2023, UW ESS - Surface event catalog, resulted in conference presentation

Nick Smoczyk (*, **) : 2022-2024, University of Minnesota - Volcano seismology using RedPy catalog, resulted in conference presentation and publication in prep

Julian Schmitt (*, +) : 2020-2021, Harvard EPS - Ambient noise seismology on cloud computing with Julia, BASIN project, resulted in conference presentation and NSF GRFP

Jared Bryan (*, **, +): 2019, Harvard EPS - Ambient noise monitoring, SCEC Internship, resulted in conference presentation, publication, and NSF GRFP

Albert Aguilar (*): 2018, Harvard EPS - Subduction zone seismology and data mining, SCEC, resulted in conference presentation

Leore Lavin: 2016, Harvard EPS - Ground motion simulations with ambient noise

Roy Bowling (*): 2014, Scripps IGPP - Ambient noise seismology, SCEC Internship

Tara Larrue (*): 2012, Stanford Geophysics - Ambient noise seismology, SURGE program

Penprapa Wutthijuk (*): 2011, Stanford - Ambient noise seismology, SURGE program

Training Workshops

- 2025 [CRESCENT-SCOPED Workshop](#) co-PI organizer for ML and Earthquake Catalog Building workshop (35 participants)
- 2024 [SCOPED Workshop](#) Lead PI organizer for Cloud/HPC/wavefield simulations workshop (100 participants)
- 2024 [SSA Cloud 101 & Data Mining](#) Lead organizer of cloud workshop (80 participants)
- 2023 [CyberTraining Workshop](#) Lead PI, coordinator, and instructor for HPC and Data Science
- 2021 – present [GeoSMART ML for Solid Earth Science](#) Co-organizer of ML training with Jupyter Book
- 2016 **SCEC-ERI VISES Summer School** Scientific committee member and instructor

University Service

- 2026 – present **Department Colloquium Committee**, University of Washington
 - Help organizing the department colloquium series
- 2026 – present **AI@UW Advisory Committee to the VP**, University of Washington
 - Advising AI strategy and initiatives across the university, commitment ~ 1hr/week

- 2024 – present **Co-director of UW CS4Env**, University of Washington
- Computer Science for the Environment (CS4Env) is a cross-unit initiative to promote synergies between computer science and environmental research at UW, includes organizing biweekly seminar, reviewing proposals, and organizing an annual symposium.
- 2024 – present **Chair of Curriculum Committee**, UW/ESS
- Reviewing proposed courses, course changes, undergraduate and graduate program.
- 2022 – present **Member of Curriculum Committee and Data Science Oversight Committee**, UW/ESS
- Review the proposed data science option program
- 2022 – 2023 **Member of Executive Committee**, UW/ESS
- 2022 – 2023 **Member of Research Faculty Search Committees**, UW/ESS
- Search committee for PNSN Network Manager and Igneous Processes Faculty Search
- 2016 – 2020 **Multiple Committee Memberships**, Harvard University
- Undergraduate Curriculum Committee, Graduate Student Council, IT Committee, Diversity Inclusion and Belonging Committee, Department Colloquium Committee

Professional Service

- 2026 – present **Member of Board of Directors at Earthscope Consortium**,
- 2024 – 2025 **Member of Science Steering Committee at SCEC**,
- 2023 – 2025 **Chair of Integration and Innovation Advisory Committee at Earthscope Consortium**,
- 2022 – 2022 **Committee Member of Charles Richter Early Career Award Committee at Seismological Society of America**,
- 2021 – 2022 **Member of Data Service Standing Committee at IRIS**,
- 2021 – 2023 **Member of HPC Standing Committee at SCEC**,
- 2011 – 2012 **Chair of Graduate Student Council at Stanford University**,

Review Panels

- 2020,2024 **NSF Review Panels EAR-Geophysics and GeoInformatics**,
- Reviewed 10+ proposals per panel
- 2016 – 2018 **USGS Review Panels Earthquake Hazards Program**,
- Reviewed 10+ proposals per panel

Editorial Service

- 2025 – present **Editor at Geophysical Journal International**,
- 2017 – 2020 **Associate Editor at Geophysical Research Letters**,
- 2014 – present **Reviewer for multiple journals in geophysics and seismology**,
- >150 reviews for GJI, BSSA, Nature Communications, GRL, JGR, Science, and others

Publications Legend

Author Notation: **Bold:** Marine A. Denolle (PI). *Italic:* graduate students, postdocs, and undergraduate students mentored by MD.

Publications

- 2026 **Distributed Acoustic Sensing Records of Earthquakes and Surface Processes at Mount Rainier Volcano**
Veronica Gaete-Elgueta, *Manuela Köpfli*, Dominik Gräff, Bradley P Lipovsky, **Marine A. Denolle**, Weston A. Thelen, Brendan Pratt, Taylor R. Kenyon
(in review in Seismica)
- 2026 **Active prot thrusts and fluid highways: Seismic noise reveals hidden subduction dynamics in Cascadia**
Maleen Kidiwela, **Marine A. Denolle**, William S. D. Wilcock, *Kuan-Fu Feng*
[10.1126/sciadv.aea3684](https://doi.org/10.1126/sciadv.aea3684) (Science Advances, 12)
- 2026 **DeepSubDAS: an earthquake phase picker from submarine distributed acoustic sensing data**
Han Xiao, Frederik Tilmann, Martijn van den Ende, Diane Rivet, Afonso Loureiro, Takeshi Tsuji, Arantza Ugalde, *Qibin Shi*, **Marine A Denolle**
[10.1093/gji/ggag061](https://doi.org/10.1093/gji/ggag061) (Geophysical Journal International, 245)
- 2026 **Exploration of Machine Learning Methods to Seismic Event Discrimination in the Pacific Northwest**
Akash Kharita, **Marine Denolle**, Alexander Hutko, Renate Hartog, Stephen Malone
[10.26443/seismica.v5i1.2068](https://doi.org/10.26443/seismica.v5i1.2068) (Seismica, 5)
- 2026 **Seismological Analysis of Contemporary and Future Alpine Fault Earthquakes Using the Southern Alps Long Skinny Array (SALSA)**
John Townend, Ilma del Carmen Juarez-Garfias, Olivia Pita-Sllim, Calum J. Chamberlain, Emily Warren-Smith, Caroline Holden, Kasper van Wijk, **Marine Denolle**, Andrew Curtis, Hiroe Miyake
[10.1785/0220250260](https://doi.org/10.1785/0220250260) (Seismological Research Letters)
- 2026 **Probing the Seattle Basin Edge Using a Dense Urban Nodal Array in 100 Backyards**
Natasha Toghramdian, **Marine A. Denolle**, *Laura Ermer*, *Chengxin Jiang*
[10.1785/0220250241](https://doi.org/10.1785/0220250241) (Seismological Research Letters)
- 2026 **The Seismic Wavefield Common Task Framework**
Alexey Yermakov, Yue Zhao, **Marine Denolle**, *Yiyu Ni*, Philippe M. Wyder, Judah Goldfeder, Stefano Riva, *Jan Williams*, David Zoro, Amy Sara Rude, Matteo Tomasetto, Joe Germany, Joseph Bakarji, Georg Maierhofer, Miles Cranmer, J. Nathan Kutz
[10.48550/arXiv.2512.19927](https://doi.org/10.48550/arXiv.2512.19927) (arXiv)
- 2026 **A Decadal Survey of the Near-Surface Seismic Velocity Response to Hydrological Variations in Utah, United States**
Kuan-Fu Feng, **Marine Denolle**, Fan-Chi Lin, Tonie van Dam
(Journal of Geophysical Research: Solid Earth, 131)
- 2025 **A Global-scale Database of Seismic Phases from Cloud-based Picking at Petabyte Scale**
Yiyu Ni, **Marine Denolle**, Amanda Thomas, Alex Hamilton, Jannes Münchmeyer, Yinzhi Wang, Loïc Bachelot, Chad Trabant, David Mencin
[10.26443/seismica.v4i2.1738](https://doi.org/10.26443/seismica.v4i2.1738) (Seismica, 4)
- 2025 **A Review of Cloud Computing and Storage in Seismology**
Yiyu Ni, **Marine A Denolle**, Jannes Münchmeyer, Yinzhi Wang, *Kuan-Fu Feng*, Carlos Garcia Jurado Suarez, Amanda M Thomas, Chad Trabant, Alex Hamilton, David Mencin
[10.1093/gji/ggaf322](https://doi.org/10.1093/gji/ggaf322) (Geophysical Journal International)

- 2025 **Training the Next Generation of Seismologists: Delivering Research-Grade Software Education for Cloud and HPC Computing Through Diverse Training Modalities**
Marine A. Denolle, Carl Tape, Ebru Bozdağ, Yinzhi Wang, Felix Waldhauser, Alice-Agnes Gabriel, Jochen Braunmiller, Bryant Chow, Liang Ding, *Kuan-Fu Feng*, Ayon Ghosh, Nathan Groebner, Aakash Gupta, *Zoe Krauss*, Amanda M. McPherson, Masaru Nagaso, Zihua Niu, *Yiyu Ni*, Ridvan Örsvuran, Gary Pavlis, Felix Rodriguez-Cardozo, Theresa Sawi, David Schaff, Nico Schliwa, David Schneller, *Qibin Shi*, Julien Thurin, Chenxiao Wang, Kaiwen Wang, Jeremy Wing Ching Wong, Sebastian Wolf, *Congcong Yuan*
[10.1785/0220240413](https://doi.org/10.1785/0220240413) (Seismological Research Letters, 96)
- 2025 **Ambient field seismology in critical zone hydrological sciences**
Marine A. Denolle, *Qibin Shi*, *Tim Clements*, *Loïc Viens*, Veronica Rodriguez-Tribaldos, Fabrice Cotton
[10.5802/crgeos.310](https://doi.org/10.5802/crgeos.310) (Comptes Rendus. Géoscience, 357)
- 2025 **Denoising Offshore Distributed Acoustic Sensing Using Masked Auto-Encoders to Enhance Earthquake Detection**
Q. Shi, **M. A. Denolle**, *Y. Ni*, *E. F. Williams*, N. You
[10.1029/2024JB029728](https://doi.org/10.1029/2024JB029728) (JGR: Solid Earth, 130)
- 2025 **Multiplexed Distributed Acoustic Sensing Offshore Central Oregon**
Q. Shi, *E. F. Williams*, B. P. Lipovsky, **M. A. Denolle**, W. S. D. Wilcock, D. S. Kelley, K. Schoedl
[10.1785/0220240460](https://doi.org/10.1785/0220240460) (Seismological Research Letters, 96)
- 2024 **Wavefield reconstruction of distributed acoustic sensing: Lossy compression, wavefield separation, and edge computing**
Y. Ni, **M. A. Denolle**, *Q. Shi*, B. P. Lipovsky, S. Pan, J. N. Kutz
[10.1029/2024JH000247](https://doi.org/10.1029/2024JH000247) (Journal of Geophysical Research: Machine Learning and Computation, 1)
- 2024 **Analysing Volcanic, Tectonic, and Environmental Influences on the Seismic Velocity from 25 Years of Data at Mount St. Helens**
P. Makus, **M. A. Denolle**, C. Sens-Schönfelder, *M. Köpfli*, F. Tilmann
[10.1785/0220240088](https://doi.org/10.1785/0220240088) (Seismological Research Letters, 95)
- 2024 **Examining 22 Years of Ambient Seismic Wavefield at Mount St. Helens**
M. Köpfli, **M. A. Denolle**, W. Thelen, P. Makus, S. Malone
[10.1785/0220240079](https://doi.org/10.1785/0220240079) (Seismological Research Letters, 95)
- 2024 **Impact of Temperature and Relative Humidity Variations on Coda Waves in Concrete**
F. Diewald, **M. Denolle**, J. J. Timothy, C. Gehlen
[10.1038/s41598-024-69564-4](https://doi.org/10.1038/s41598-024-69564-4) (Scientific Reports, 14)
- 2024 **Monitoring velocity change over 20 years at Parkfield**
K. Okubo, B. Delbridge, **M. Denolle**
[10.1029/2023JB028084](https://doi.org/10.1029/2023JB028084) (Journal of Geophysical Research: Solid Earth, 129)
- 2024 **Extended crack propagation by local nucleation and rapid transverse expansion**
T. Cochar, I. Svetlizky, G. Albertini, R. C. Viesca, S. M. Rubinstein, F. Spaepen, *C. Yuan*, **M. Denolle**,
Y.-Q. Song, L. Xiao, D. A. Weitz
[10.1038/s41567-023-02365-0](https://doi.org/10.1038/s41567-023-02365-0) (Nature Physics)
- 2023 **Discrimination between icequakes and earthquakes in southern Alaska: an exploration of waveform features using random forest algorithm**
A. Kharita, **M. Denolle**, M. West
[10.1093/gji/ggae106](https://doi.org/10.1093/gji/ggae106) (Geophysical Journal International)
- 2023 **Ocean Coupling Limits Rupture Velocity of Fastest Observed Ice Shelf Rift Propagation Event**
S. Olinger, B. Lipovsky, **M. Denolle**
[10.1029/2023AV001023](https://doi.org/10.1029/2023AV001023) (AGU Advances, 5)

- 2023 **Laboratory hydrofracture as analogs to tectonic tremors**
C. Yuan, T. Cochard, **M. Denolle**, J. Gombert, A. Wech, L. Xiao, D. Weitz
[10.1029/2023AV001002](https://doi.org/10.1029/2023AV001002) (AGU Advances, 5)
- 2023 **Improved observations of deep earthquake ruptures using machine learning**
Q. Shi, **M. Denolle**
[10.1029/2023JB027334](https://doi.org/10.1029/2023JB027334) (Journal of Geophysical Research: Solid Earth, 128)
- 2023 **Better Together: Ensemble Learning for Earthquake Detection and Phase Picking**
Congcong Yuan, Yiyu Ni, Youzuo Lin, **Marine Denolle**
[10.1109/TGRS.2023.3320148](https://doi.org/10.1109/TGRS.2023.3320148) (IEEE Transactions on Geoscience and Remote Sensing, 61)
- 2023 **An Object Storage for Distributed Acoustic Sensing**
Yiyu Ni, **Marine A. Denolle**, Rob Fatland, Naomi Alterman, Bradley P. Lipovsky, Friedrich Knuth
[10.1785/0220230172](https://doi.org/10.1785/0220230172) (Seismological Research Letters, 95)
- 2023 **Seismology in the cloud: guidance for the individual researcher**
Z. Krauss, Y. Ni, S. Henderson, **M. Denolle**
[10.26443/seismica.v2i2.979](https://doi.org/10.26443/seismica.v2i2.979) (Seismica, 2)
- 2023 **Curated Pacific Northwest AI-ready Seismic Dataset**
Y. Ni, A. Hutko, F. Skene, **M. Denolle**, S. Malone, P. Bodin, R. Hartog, A. Wright
[10.26443/seismica.v2i1.368](https://doi.org/10.26443/seismica.v2i1.368) (Seismica, 2)
- 2023 **Probing environmental and tectonic changes underneath Ciudad de México with the urban seismic field**
L. Ermert, E. Cabral-Cano, E. Chaussard, D. Solano-Rojas, L. Quintanar, D. Morales Padilla, E. A. Fernandez-Torres, **M. A. Denolle**
[10.5194/egusphere-2022-1361](https://doi.org/10.5194/egusphere-2022-1361) (Solid Earth (EGU))
- 2023 **The Seismic Signature of California's Earthquakes, Droughts, and Floods**
T. Clements, **M. A. Denolle**
[10.1029/2022JB025553](https://doi.org/10.1029/2022JB025553) (Journal of Geophysical Research: Solid Earth, 128)
- 2022 **A multitask encoder–decoder to separate earthquake and ambient noise signal in seismograms**
J. Yin, **M. A. Denolle**, B. He
[10.1093/gji/ggac290](https://doi.org/10.1093/gji/ggac290) (Geophysical Journal International, 231)
- 2022 **Pronounced Seismic Anisotropy in Kanto Sedimentary Basin: A Case Study of Using Dense Arrays, Ambient Noise Seismology, and Multi-Modal Surface-Wave Imaging**
C. Jiang, **M. A. Denolle**
[10.1029/2022JB024613](https://doi.org/10.1029/2022JB024613) (Journal of Geophysical Research: Solid Earth, 127)
- 2022 **Imaging the Kanto Basin seismic basement with earthquake and noise autocorrelation functions**
L. Viens, C. Jiang, **M. A. Denolle**
[10.1093/gji/ggac101](https://doi.org/10.1093/gji/ggac101) (Geophysical Journal International, 230)
- 2022 **Tracking the Cracking: a Holistic Analysis of Rapid Ice Shelf Fracture Using Seismology, Geodesy, and Satellite Imagery on the Pine Island Glacier Ice Shelf, West Antarctica**
S. D. Olinger, B. P. Lipovsky, **M. A. Denolle**, B. W. Crowell
[10.1029/2021GL097604](https://doi.org/10.1029/2021GL097604) (Geophysical Research Letters)
- 2022 **Detecting Elevated Pore Pressure due to Wastewater Injection Using Ambient Noise Monitoring**
Z. Yang, C. Yuan, **M. A. Denolle**
[10.1785/0320210036](https://doi.org/10.1785/0320210036) (The Seismic Record, 2)

- 2021 **The Earth's Surface Controls the Depth-Dependent Seismic Radiation of Megathrust Earthquakes**
J. Yin, M. A. Denolle
[10.1029/2021AV000413](https://doi.org/10.1029/2021AV000413) (AGU Advances, 2)
- 2021 **Comparing approaches to measuring seismic phase variations in the time, frequency, and wavelet domains**
C. Yuan, J. Bryan, M. A. Denolle
[10.1093/gji/ggab140](https://doi.org/10.1093/gji/ggab140) (Geophysical Journal International, 226)
- 2021 **Source time function clustering reveals patterns in earthquake dynamics**
J. Yin, Z. Li, M. A. Denolle
[10.1785/0220200403](https://doi.org/10.1785/0220200403) (Seismological Research Letters, 92)
- 2021 **SeisNoise.jl: Ambient Seismic Noise Cross Correlation on the CPU and GPU in Julia**
T. Clements, M. A. Denolle
[10.1785/0220200192](https://doi.org/10.1785/0220200192) (Seismological Research Letters, 92)
- 2020 **Quiet Anthropocene, quiet Earth**
M. A. Denolle, T. Nissen-Meyer
[10.1126/science.abd8358](https://doi.org/10.1126/science.abd8358) (Science, 369)
- 2020 **SeisIO: A Fast, Efficient Geophysical Data Architecture for the Julia Language**
J. P. Jones, K. Okubo, T. Clements, M. A. Denolle
[10.1785/0220190295](https://doi.org/10.1785/0220190295) (Seismological Research Letters, 91)
- 2020 **NoisePy: A new high-performance python tool for ambient-noise seismology**
C. Jiang, M. A. Denolle
[10.1785/0220190364](https://doi.org/10.1785/0220190364) (Seismological Research Letters, 91)
- 2019 **Earthquakes Within Earthquakes: Patterns in Rupture Complexity**
P. Danré, J. Yin, B. Lipovsky, M. Denolle
[10.1029/2019GL083093](https://doi.org/10.1029/2019GL083093) (Geophysical Research Letters, 43)
- 2019 **Long-period ground motions from past and virtual megathrust earthquakes along the Nankai Trough, Japan**
L. Viens, M. Denolle
[10.1785/0120180320](https://doi.org/10.1785/0120180320) (Bulletin of the Seismological Society of America, 109)
- 2019 **Energetic Onset of Earthquakes**
M. Denolle
[10.1029/2018GL080687](https://doi.org/10.1029/2018GL080687) (Geophysical Research Letters, 46)
- 2019 **Relating teleseismic backprojection images to earthquake kinematics**
J. Yin, M. Denolle
[10.1093/gji/ggz048](https://doi.org/10.1093/gji/ggz048) (Geophysical Journal International, 217)
- 2019 **Geometric Controls on Pulse-like Rupture in a Dynamic Model of the 2015 Gorkha Earthquake**
Y. Wang, M. Denolle, S. M. Day
[10.1029/2018JB016602](https://doi.org/10.1029/2018JB016602) (Journal of Geophysical Research, 124)
- 2018 **Tracking ground water using the ambient seismic field**
T. Clements, M. Denolle
[10.1029/2018GL077706](https://doi.org/10.1029/2018GL077706) ((User text suggests possible mismatch of volume/issue) Geophysical Research Letters, 123)
- 2018 **Complex near-surface rheology inferred from the response of greater Tokyo to strong ground motions**
L. Viens, M. Denolle, N. Hirata, S. Nakagawa
 (Journal of Geophysical Research: Solid Earth, 123)

- 2018 **Strong Shaking Predicted in Tokyo From an Expected M7+ Itoigawa-Shizuoka Earthquake**
M. A. Denolle, P. Boué, N. Hirata, G. C. Beroza
 (Journal of Geophysical Research: Solid Earth, 123)
- 2018 **Improved model fitting for the empirical Green's function approach using hierarchical models**
C. Van Houtte, **M. Denolle**
 (Journal of Geophysical Research: Solid Earth, 123)
- 2018 **Tracking groundwater levels using the ambient seismic field**
T. Clements, **M. A. Denolle**
[10.1029/2018GL077706](https://doi.org/10.1029/2018GL077706) (Geophysical Research Letters, 45)
- 2018 **Convolutional neural network for earthquake detection and location**
T. Perol, M. Gharbi, **M. Denolle**
 (Science Advances, 4)
- 2018 **Spatial and Temporal Evolution of Earthquake Dynamics: Case Study of the Mw 8.3 Illapel Earthquake, Chile**
J. Yin, **M. A. Denolle**, H. Yao
[10.1002/2017JB014265](https://doi.org/10.1002/2017JB014265) (Journal of Geophysical Research: Solid Earth, 123)
- 2017 **Multicomponent C3 Green's Functions for Improved Long-Period Ground-Motion Prediction**
 Y. Sheng, **M. A. Denolle**, G. C. Beroza
[10.1785/0120170053](https://doi.org/10.1785/0120170053) (Bulletin of the Seismological Society of America, 107)
- 2017 **Retrieving impulse response function amplitudes from the ambient seismic field**
L. Viens, **M. Denolle**, H. Miyake, S. Sakai, S. Nakagawa
[10.1093/gji/ggx155](https://doi.org/10.1093/gji/ggx155) (Geophysical Journal International, 210)
- 2016 **Beyond Basin Resonance: Characterizing Wave Propagation Using a Dense Array and the Ambient Seismic Field**
 P. Boue, **M. Denolle**, N. Hirata, S. Nakagawa, G. C. Beroza
[10.1093/gji/ggw205](https://doi.org/10.1093/gji/ggw205) (Geophysical Journal International, 206)
- 2016 **New perspective on self-similarity of shallow thrust earthquakes**
M. Denolle, P. M. Shearer
[10.1002/2016JB013105](https://doi.org/10.1002/2016JB013105) (Journal of Geophysical Research: Solid Earth, 121)
- 2015 **Dynamics of the M7.8 2015 Nepal Earthquake**
M. Denolle, W. Fan, P. M. Shearer
[10.1002/2015GL065336](https://doi.org/10.1002/2015GL065336) (Geophysical Research Letters, 42)
- 2014 **Full 3D Tomography (F3DT) for Crustal Structure in Southern California Based on the Scattering-Integral (SI) and the Adjoint-Wavefield (AW) Methods**
 E.-J. Lee, P. Chen, T. H. Jordan, P. B. Maechling, **M. Denolle**, G. C. Beroza
[10.1002/2014JB011236](https://doi.org/10.1002/2014JB011236) (Journal of Geophysical Research, 119)
- 2014 **Long-period seismic amplification in the Kanto Basin from the ambient seismic field**
M. Denolle, H. Miyake, S. Nakagawa, N. Hirata, G. C. Beroza
[10.1002/2014GL059425](https://doi.org/10.1002/2014GL059425) (Geophysical Research Letters, 41)
- 2014 **Strong Ground Motion Prediction using Virtual Earthquakes**
M. Denolle, E. M. Dunham, G. A. Prieto, G. C. Beroza
[10.1126/science.1245678](https://doi.org/10.1126/science.1245678) (Science, 343)
- 2013 **Ground Motion Prediction of Realistic Earthquake Sources Using the Ambient Seismic Field**
M. Denolle, E. M. Dunham, G. A. Prieto, G. C. Beroza
[10.1029/2012JB009603](https://doi.org/10.1029/2012JB009603) (Journal of Geophysical Research, 118)

- 2013 **A numeric evaluation of attenuation from ambient noise correlation functions**
J. F. Lawrence, **M. Denolle**, K. J. Seats, G. Prieto
[10.1002/2012JB009513](#) (Journal of Geophysical Research, 118)
- 2012 **Solving the Surface-Wave Eigenproblem with Chebyshev Spectral Collocation**
M. Denolle, E. M. Dunham, G. C. Beroza
[10.1785/0120110183](#) (Bulletin of the Seismological Society of America, 102)
- 2011 **On amplitude carried by the ambient seismic field**
G. A. Prieto, **M. Denolle**, J. F. Lawrence, G. C. Beroza
(Comptes Rendus Geoscience (Thematic Issue: Imaging and Monitoring with Seismic Noise), 343)
- 2010 **Evidence of active backthrusting at the NE Margin of Mentawai Islands, SW, Sumatra**
S. Singh, N. Hananto, A. Chauhan, H. Permana, **M. Denolle**, A. Hendriyana, D. Natawidjaja
[10.1111/j.1365-246X.2009.04458.x](#) (Geophysical Journal International, 180)
- Agroseismology and the impact of farming practices on soil hydrodynamics**
Qibin Shi, David R. Montgomery, Abigail L.S. Swann, Nicoleta C. Cristea, *Ethan F. Williams*, Nan You, Simon Jeffery, Joe Collins, Ana Prada Barrio, Paula A. Misiewicz, Tarje Nissen-Meyer, **Marine A. Denolle**
[10.1126/science.aec0970](#) (Science, 0)

Grants

- 2025 **Jerry and Linda Paros** Multi-Sensor, Multi-Geohazards Monitoring at Mt Rainier and Mountaineous regions: \$750,000 (lead PI)
- 2025 **UW CRESST - FFST** Toward Predictive Understanding of Climate-Compounded Geohazards: \$300,000 (lead PI)
- 2025 **NSF CS4All** Separating the Signal from the Noise: Promoting Alaskan students inquiry with geographically relevant seismic data and machine learning techniques: \$889,766 (lead PI Lore, Denolle co-PI, \$62,000 to UW), DRL-2524060
- 2025 **NSF RISE CAIG** Collaborative Research: CAIG: Framework for Artificial Intelligence-Enhanced Modeling of Wildfire Geohazards (FAIM-WG): Applications for postfire debris flows across the Western US: \$735,074 (lead PI Istanbuluoglu, Denolle co-PI), RISE-2530591
- 2025 **NSF OCE** Multi-span distributed fiber sensing on the Ocean Observatories Initiative Regional Cabled Array: \$880,490 (lead PI Wilcock, Denolle co-PI), OCE-2526198, technology development grant.
- 2025 **NSF OCE** EXTension on the ENDeavour Segment (EXTEND): Illuminating the seafloor spreading cycle: \$785,819.00 (lead PI Wilcock, Denolle co-PI), OCE-2439442
- 2024 **NSF EAR** Collaborative Research: A seismic investigation of slow slip and fault locking along the Alaska-Aleutian subduction zone: \$248,835.00 (Denolle co-PI, collaborative with lead PI Golos), EAR-2346079
- 2023 **NSF OCE** RAPID: Multiplexed Distributed Acoustic Sensing (DAS) at the Ocean Observatory Initiative (OOI) Regional Cabled Array (RCA): \$198,069.00 (lead PI Wilcock, Denolle co-PI), OCE-2415521
- 2023 **Ecole Normal Supérieure de Paris** Travel Fellowship: \$3,500 (Denolle PI)
- 2023 **SCEC** CyberTraining for Seismology: Data Science and HPC: \$35,229 (Denolle PI)
- 2021 **David and Lucile Packard Foundation** URG Program: \$50,000 (PI)

- 2021 **Murdock Charitable Trust Fund** UW FiberLab equipment: \$950,000 (co-PI, lead PI Brad Lipovsky). Built facility for Fiber Sensing.
- 2021 **NSF CyberTraining** GeoSMART ML workforce development: \$995,817 (co-PI). Build a course MLGEO.
- 2021 **NSF OAC-CSSI** Collaborative Research: Frameworks: Seismic COmputational Platform for Empowering Discovery (SCOPED): \$717,441.00 (Denolle PI, lead-PI Carl Tape), OAC-2103701
- 2021 **NSF EAR, PREEVENTS** Collaborative Research: Cross-Validation of Empirical and Physics-based ground motion predictions: \$167,804.00 (Denolle PI), EAR-2125337
- 2020 **SCEC grant** Aftershock patterns and co-seismic off-fault damage elucidate dynamic rupture processes on the 2019 Ridgecrest earthquake sequence: \$33,307 (Denolle PI) - returned
- 2019 **Harvard University - David Rockefeller Center for Latin American Studies** Monitoring Seismic Hazards in Mexico City using Grillo, a Low-Cost Earthquake Early Warning System, Hardware, \$85,000 (Denolle PI)
- 2019 **Harvard Data Science Initiative** Ambient-noise seismology using Cloud Computing: \$27,210 (Denolle PI)
- 2018 **NSF EAR, CAREER** CAREER: Dynamics of surface rupturing thrust earthquakes: \$504,315 (PI)
- 2018 **SCEC grant** Data Collection for Virtual Earthquakes on Cajon Pass: \$28,085 (Denolle PI) - field work
- 2017 **David and Lucile Packard Foundation** Fellowship: \$875,000 (PI)
- 2017 **NSF PREEVENTS** Collaborative Proposal - PREEVENTS Track 2: Cascadia Scenario Earthquakes: Source, Path, and implications for Earthquake Early Warning: \$324,495.00 (Denolle PI, collaborative with lead PI Yihe Huang and Amanda Thomas), EAR-1739712
- 2017 **SCEC grant** Static and dynamic source parameters of global strike-slip earthquakes: \$25,000 (Denolle PI)
- 2016 **SCEC grant** Epistemic uncertainties in ground motion prediction from virtual earthquakes: \$26,173 (Denolle PI)
- 2015 **SCEC grant** Basin Response to Virtual Earthquakes on the San Jacinto Fault and the Itoigawa-Shizuoka Fault: \$20,000 (Denolle PI)

Research Products

- 2025 **Global Seismic Phase Database** Dataset
Global seismic data product from cloud-scale waveform processing. [DOI](#).
- Scale: 4.3 billion phase picks from 1.3 PB of continuous seismic data.
 - Publication link: Seismica (2025) with DOI-backed archival record.
 - Supports global monitoring and data-driven geophysics workflows.
- 2023 **EarthML4PNW / PNW AI-Ready Seismic Dataset** Dataset
Curated ML-ready Pacific Northwest dataset product. [Dataset repository](#).
- Publication link: Seismica (2023), DOI [10.26443/seismica.v2i1.368](#).
 - Version-controlled release and metadata stewardship through GitHub workflows.
 - Designed for reproducible benchmarking and graduate-level ML coursework.

- 2021 – present **Machine Learning in the Geosciences (Open-Access Jupyter Book)** Textbook
[Book](#) and [course repository](#) provide graduate-level ML curriculum, asynchronous modules, and home-work data pipelines for geoscience training.
- Addresses a core gap: no canonical ML textbook existed for geoscience graduate classrooms.
 - Adoption and contribution interest from peer programs (including Arizona and Berkeley).
 - Ongoing community development with curated datasets and reproducible teaching workflows.
- 2020 – present **SeisNoise.jl** Software
 Open-source Julia software for high-performance ambient-noise processing and cross-correlation.
[GitHub](#).
- Extends the Julia seismology ecosystem with core ambient-noise tooling.
 - Community signal: 50 stars and 17 forks (Apr 2023 snapshot).
 - Used by group members for production research workflows.
- 2019 – present **NoisePy** Software
 Open-source Python software for large-scale ambient-noise seismology. [GitHub](#).
- Community scale: 207 stars, 83 forks, 19 contributors (Jan 2026).
 - Operational usage: taught in workshops and used for large-scale cross-correlation workflows.
 - Open, version-controlled development supporting reproducible and cloud-oriented processing.

Invited Talks

- 2026 **NGF Science Meeting** Conference Speaker
- 2026 **WCCM 2026, Munich** Keynote Speaker
- 2026 **European Geophysical Union** Conference Speaker
- 2026 **CERI Memphis** Department Colloquium
- 2025 **IEEE New Era AI World Leader Summit, Bothell** Keynote Speaker
- 2025 **SIAM Geoscience Meeting** Plenary Speaker
- 2025 **Workshop on Earthquake Physics and Applications of AI to Tectonic Faulting, Italy** Plenary Speaker
- 2025 **Civil Environmental Engineering, University of Washington** Department Colloquium
- 2025 **Radcliffe Institute of Advanced Studies, on the road** Short Talk
- 2025 **Washington University, Saint Louis** Department Colloquium
- 2024 **University of Southern California** Department Colloquium
- 2024 **University of California, Davis** Department Colloquium
- 2024 **Northern Arizona University** Department Colloquium
- 2024 **Passive imaging and monitoring in wave physics (Cargese, France)** Invited Conference Talk
- 2024 **Statewide California Earthquake Center** Plenary Speaker
- 2023 **Ecole Normale Supérieure, Paris Séminaire** Departemental

2023 **eScience Institute, University of Washington** Data Science Seminar

2023 **Sandia National Lab, GNEM seminar series** Department Colloquium

2022 **American Geophysical Union (x2)** Invited Conference Talk

2022 **University of New Mexico** Department Colloquium

2021 **University of Oregon** Seismo Colloquium

2021 **U Utah, Seismo Tea** Seismo Colloquium

2021 **University of Wisconsin** Department Colloquium

2021 **Colorado School of Mines** Department Colloquium

2021 **Mexico a traves de los sismos** Invited Conference Talk

2020 **University of Washington** Department Colloquium

2020 **UC Berkeley** Department Colloquium

2019 **Yale University** Department Colloquium

2019 **University of Washington, seismolunch** Seismo Colloquium

2019 **EGU annual meeting, Vienna** Invited Conference Talk

2019 **Michigan State University** Department Colloquium

2019 **Victoria University of Wellington, SN Jepson Lecture, New Zealand** Public Lecture

2019 **GNS-Science, New Zealand** Department Colloquium

2019 **Stanford University, Department of Geophysics** Department Colloquium

2019 **Tufts University, Civil Engineering seminar** Department Colloquium

2018 **Brown University** Department Colloquium

2018 **Ecole Normale Supérieure, Paris** Department Colloquium

2017 **AGU, New Orleans** Invited Conference Talk

2017 **Columbia University, Lamont Doherty Earth Observatory** Department Colloquium

2016 **Harvard Museum of Natural History** Public Lecture

2016 **University of Oregon** Department Colloquium

2016 **University of New Hampshire, Chapman Colloquium** Department Colloquium

2016 **UC Santa Cruz, IGPP seminar** Department Colloquium

2016 **Massachusetts Institute of Technology** Department Colloquium

- 2015 **USGS, Menlo Park, Earthquake Hazard Program** Department Colloquium
- 2015 **University of Victoria, BC, Canada** Department Colloquium
- 2015 **Penn State, Geodynamics seminar** Department Colloquium
- 2015 **Harvard, Earth and Planetary Sciences** Department Colloquium
- 2015 **UT Austin, Solid Earth seminar** Department Colloquium
- 2015 **UCLA, seismology/tectonics seminar** Department Colloquium
- 2015 **HOKUDAN International Symposium on Active Faulting, Awaji, Japan** Invited Conference Talk
- 2015 **Information Theory and Applications workshop, La Jolla** Invited Conference Talk
- 2015 **IGPP-Scripps Institution of Oceanography, UCSD, Geophysics seminar** Department Colloquium
- 2015 **University of Southern California, Geophysics seminar** Department Colloquium
- 2014 **Strong Motion Studies for Future Mega-Quakes, DPRI, Kyoto University, Japan** Invited Conference Talk
- 2014 **AGU-SEG Summer Research workshop, Vancouver, Canada** Invited Conference Talk
- 2014 **San Diego State University** Department Colloquium
- 2014 **UC Santa Barbara** Department Colloquium
- 2014 **IGPP-Scripps Institution of Oceanography, UCSD, Geophysics seminar** Department Colloquium
- 2013 **AGU, Meeting of the Americas, Cancun, Mexico** Invited Conference Talk
- 2013 **Caltech Seismo Lab** Department Colloquium
- 2013 **USGS, Menlo Park** Department Colloquium
- 2013 **Stanford ICME seminar** Department Colloquium
- 2013 **Earthquake Research Institute, Tokyo University, Japan** Department Colloquium
- 2013 **Advanced Industrial Science and Technology, Japan** Department Colloquium
- 2013 **Disaster Prevention Research Institute, Japan** Department Colloquium
- 2012 **ACOUSTICS, France** Invited Conference Talk
- 2012 **Berkeley Seismo Lab** Department Colloquium
- 2011 **Institut de Physique du Globe de Paris, Earthquake seminar** Department Colloquium